Chapter 10

Soil Microbiology

Chapter Outline

10.1 Soil in General

10.2 Pioneers of Soil Microbiology

10.3 Components of Soil

10.4 Soil Microorganisms

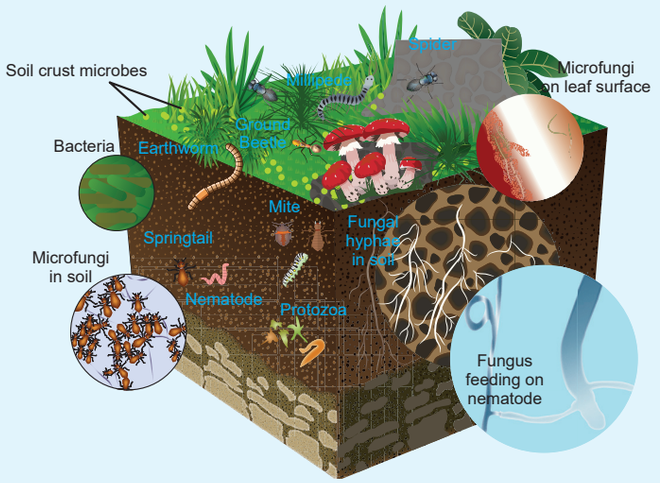
10.5 Microbial Interactions

10.6 Rhizosphere

10.7 Phyllospere

10.8 Spermosphere

ESFIHV



Soil is inhabited by a living microscopic population which is responsible for the numerous reactions taking place in the soil. The soil microorganisms affects the life economy of man in many ways.

Learning Objectives

After studying this chapter the student will be able,

- *To know the composition of soil.*
- *To understand the importance of soil microorganisms in soil fertility.*
- *To learn about the beneficial and harmful interaction between soil microorganisms.*

Knowledge of Soil Microbiology is essential to understand the agricultural and environmental science. Without soil microorganisms, life as we know could not exist on earth. Organic matter would accumulate in the form of undecomposed substances. Why should we study soil Microbiology? If we understand what is happening in soil, we get a better idea of how other biological systems work on earth.

10.1 Soil in General

Soil is the outer covering of the earth. It consists of loosely arranged layer of materials composed of inorganic and organic constituents. Soil provides the physical support needed for the anchorage of root system and serves as the reservoir of air, water and nutrients that are essential for plant growth.

10.1.1 Formation of Soil

The processes involved in the formation of soil are slow, gradual and continuous. The sum total of environmental effects on rocks collectively known as the weathering of rocks. Weathering of rocks is a continuous phenomenon and add more and more soil to the surface of the earth. There are different types of parent materials of rocks available for the formation of soil.